Parental Altruism and Transfers

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Abstract

Using data from the Health and Retirement Study (1998–2018), I show that child incomes, parent incomes, and parent assets, together with additional observables, explain less than 20 percent of the variation in inter vivos financial transfers. This finding is inconsistent with standard models of homogeneous altruism, in which transfer variation is driven entirely by differences in parent and child incomes and assets. I develop a static model of heterogeneous parental altruism in which altruism follows a lognormal distribution. The model generates transfers across parent and child income distributions, including positive transfers to relatively high-income children, a feature absent from standard homogeneous-altruism models. These distributional implications are important for policy evaluation, as standard models may overstate familial support for low-income children while understating support for high-income children.

Keywords: intergenerational transfers, parental altruism, informal insurance

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1 Introduction

Financial transfers from parents are a major channel of intergenerational support, shaping children's consumption, wealth accumulation, and access to economic opportunity well into adulthood. In 2024, 44 percent of U.S. adults aged 18–34 reported receiving financial assistance from their parents (Fry et al., 2024). These transfers play an important role in the transmission of socio-economic status, yet the motives behind them remain only partially understood. Standard economic models typically assume homogeneous parental altruism (Altonji et al., 1997; Nishiyama, 2002), implying that variation in transfers should be explained primarily by differences in parent and child resources. However, empirical evidence consistently shows substantial heterogeneity in transfer behaviour, including positive transfers to children across the entire income distribution, even those with higher incomes than their parents. This pattern is difficult to reconcile with canonical models of homogeneous altruism, which predict that parents should direct transfers towards their lowest-income children via equating marginal utilities of consumption between themselves and their children. The observation that transfers flow broadly, rather than being targeted only to the poorest children, suggests that heterogeneity in parental motives plays a central role in shaping intergenerational support.

Using data from the Health and Retirement Study (1998–2018), I document that child incomes, parent incomes, and parent assets, along with a rich set of covariates, explain approximately 16 percent of the variation in inter vivos transfers. This limited explanatory power shows that models built on homogeneous altruism, where incomes, assets, and other observable characteristics fully determine transfer behavior, cannot match the empirical patterns. I interpret the unexplained component as reflecting unobserved heterogeneity in parental altruism and incorporate this heterogeneity into a static model in which altruism follows a lognormal distribution. Allowing altruism to vary across parents enables the model to generate transfer patterns that more closely resemble those in the data, including positive transfers to some higher-income children.

The empirical analysis draws on the Health and Retirement Study (HRS), a longitudinal survey that follows aging parents and their children in biennial waves from 1992 to the present. The HRS provides unusually rich information on demographics, income, assets, health, and detailed intergenerational transfers, making it one of the few datasets that directly observes both the incidence and magnitude of financial transfers from parents to adult children. I use the RAND HRS Family Data, which restructures the survey into roughly 150,000 parent–child pairs who are interviewed repeatedly, of which about one-third appear in the latest wave used. I restrict the main analysis to the years 1998–2018 because child income is consistently measured in comparable bins and because 2018 is the final RAND-harmonized wave before the COVID-19 pandemic. The resulting data allow me to characterize transfer behavior across the full joint distribution of parent and child resources.

Several key facts emerge from the HRS. First, higher-income parents are more likely to give, and higher-income children are less likely to receive, a transfer. Second, conditional on giving, transfer amounts are increasing in parent income and to some extent also in child income. Third, parental assets rise sharply with child income, reflecting intergenerational correlations in economic status and the strong association between income and wealth among parents.

Controlling for parent and child characteristics provides additional evidence for these patterns. I estimate an ordinary least squares model of transfer amounts on parent income, child income, parent assets, and a broad set of demographic and economic covariates. These estimates show that transfers decline sharply with child income and rise with parent income, consistent with consumption smoothing motives and the greater giving capacity of financially secure parents. Despite the richness of the HRS, this specification explains only about 16 percent of the variation in transfer amounts, highlighting substantial unexplained heterogeneity in parental transfer behaviour. A more limited specification that includes only incomes and assets explains less than 6 percent of transfer variation, reinforcing that financial resources alone cannot account for inter vivos transfers. This weak explanatory power provides direct empirical motivation for the exploration of heterogeneity in parental altruism.

Motivated by these facts, I develop a static, endowment-based model in which parents allocate resources among consumption, saving, and transfers to a single child. Parents equate the marginal utility of consuming an additional unit of their endowment with the marginal utility of increasing the child's consumption through a transfer and with the marginal value of saving. The model incorporates heterogeneity in parent and child endowments, and it allows parental altruism to vary across households.

To discipline the degree of heterogeneity in altruism, I target the share of transfer variation that can be explained by observables. Empirically, the aforementioned regression on transfer amounts can explains approximately 16 percent of the variation in transfers. I interpret the remaining unexplained component as reflecting unobserved heterogeneity in parental altruism. In the model, I estimate an analogous regression using only the variables that appear in the model and choose the variance of the altruism distribution so that this model-based regression explains the same share of variation as in the data.

Using the calibrated distribution of parental altruism, the model produces transfer patterns that differ sharply from those implied by homogeneous altruism. In the standard framework, parents transfer resources only to children with very low incomes. In contrast, the heterogeneous altruism model generates positive transfers across the child income distribution, even among children with incomes above $100,000. These findings show that the canonical theory of altruism systematically mischaracterizes private transfers, understating support to high-income children and overstating support to low-income children. Understanding the true measure of private transfers to adult children is essential in the targeting of public transfers to individuals with the greatest need.

Related Literature A large body of research examines inter vivos transfers within families, particularly those flowing from parents to adult children. Much of this literature interprets transfers through an altruistic lens, emphasizing how parents use financial support to smooth consumption across family members or to equalize living standards across generations (McGarry and Schoeni, 1995; Barro, 1974; Altonji et al., 1997). Subsequent quantitative research has focused on matching average transfer levels, explaining the joint behavior of transfers and bequests, or evaluating the extent to which private transfers offset government policy, but these frameworks have not been designed to reproduce the full distribution of transfers across the child income distribution (Nishiyama, 2002; De Nardi, 2004; Slavik and Wiseman, 2018; Barczyk, 2016). Recent work also examines heterogeneity in parental altruism in the context of post-secondary education and optimal financial aid for young adults (Park, 2025).

This paper contributes to the literature in two ways. First, it shows that standard models with homogeneous parental altruism are unable to explain the pattern of transfers observed in the data. Second, it demonstrates that introducing heterogeneity in parental altruism allows the model to better account for transfer behaviour, including positive transfers to children across the income distribution. By allowing altruism to vary across parents and interact with the joint distribution of parent income and assets, the model captures substantial variation in transfers that has remained unexplained in prior work.

The remainder of the paper proceeds as follows. Section 2 outlines the theoretical foundations of altruistic family transfers. Section 3 describes the Health and Retirement Study and presents the key empirical facts that motivate the analysis. Section 4 develops a static model of transfers from parents to children and Section 5 discusses the model estimation. Section 6 reports quantitative results for transfer flows and evaluates the model of heterogeneous altruism relative to a homogeneous baseline. Section 7 concludes.

2 Standard Theory of Parental Altruism

To formalize these ideas, consider a stylized static model in which parents and children each receive an exogenous endowment, e_p and e_k. They live for one period, face no savings decision, and the parent chooses a transfer $t \geq 0$. The parent's problem is:

$$
\begin{aligned}
&\max_{c_p, c_k, t} u(c_p) + \nu u(c_k) \\
&\text{s.t.} \quad c_p = e_p - t \\
&\qquad\quad c_k = e_k + t
\end{aligned}
$$

Under constant relative risk aversion preferences, the parent trades off the marginal utility of their own consumption against the marginal utility of their child's, scaled by the altruism parameter ν. The first-order condition equates these weighted marginal utilities and yields:

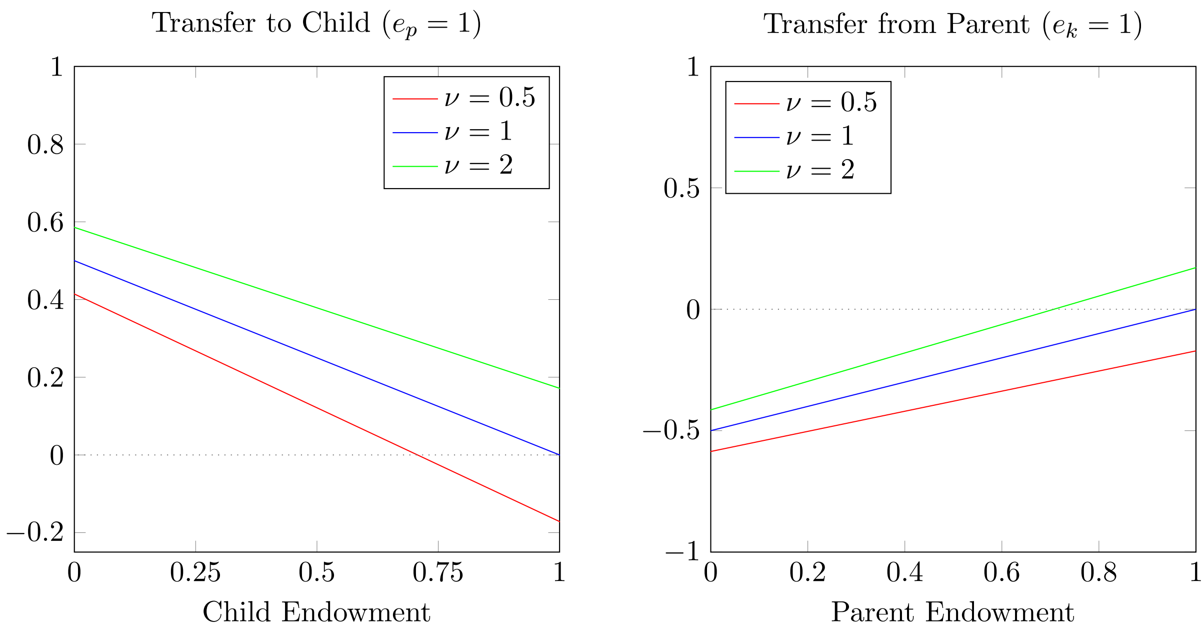
$$
\begin{aligned}
U &= \frac{(e_p - t)^{1-\sigma}}{1-\sigma} + \nu \frac{(e_k + t)^{1-\sigma}}{1-\sigma} \\
\frac{\partial U}{\partial t} &= -(e_p - t)^{-\sigma} + \nu (e_k + t)^{-\sigma} \\
\nu &= \left(\frac{e_k + t}{e_p - t} \right)^{\sigma}
\end{aligned}
$$

Solving for the optimal transfer yields:

$$
\begin{aligned}
\nu^{1/\sigma}(e_p - t) &= e_k + t \\
(1 + \nu^{1/\sigma})t &= \nu^{1/\sigma} e_p - e_k \\
t^* &= \frac{\nu^{1/\sigma} e_p - e_k}{1 + \nu^{1/\sigma}} \qquad (1)
\end{aligned}
$$

Figure 1 illustrates how transfers vary with parent and child endowments. When parents value their children's consumption at least as much as their own (ν), they give positive transfers whenever they are richer than their children. As altruism declines ($\nu < 1$), parents become more tolerant of consumption inequality and require a larger gap between their own endowment and their child's before making a positive transfer. Only highly altruistic parents ($\nu > 1$) transfer resources even when their children are relatively wealthy.

Figure 1: Transfer Functions from Parents to Children



Note: Transfers are decreasing in child endowments and increasing in parent endowments. This pattern is consistent with intergenerational consumption smoothing by altruistic parents towards their children.

Altruism can be interpreted as a parent's tolerance for consumption inequality between themselves and their children. When $\nu \leq 1$ and $e_k > e_p$ as in the second panel of Figure 1, a positive transfer actually increases inequality because the child is already better off than the parent. In fact, parents would prefer to take resources from their children if that were feasible. When $\nu > 1$, parents are willing to tolerate inequality that favours their children, and may transfer even when doing so widens the consumption gap between them.

3 Data

The Health and Retirement Study (HRS), sponsored by the National Institute on Aging and conducted by the University of Michigan, is a longitudinal panel survey that follows aging parents and their children in biennial waves from 1992 to the present. Each wave contains a nationally representative sample of roughly 20,000 individuals grouped by birth cohort. The HRS collects detailed information on demographics, health, disability, housing, employment, assets, and intergenerational transfers to parents, siblings, and children. Importantly, transfers are recorded in every wave, and characteristics of recipient children are also observed. This repeated measurement of transfers is a major advantage over comparable data sources such as the Panel Study of Income Dynamics (PSID), which is itself a long-running panel but collects information on intergenerational transfers only through infrequent cross-sectional supplements (1988 and 2013) rather than as part of the core annual survey.

My analysis uses the RAND HRS Family Data, a harmonized version of the HRS that restructures the raw survey into parent–child pairs observed repeatedly over time. The full dataset contains roughly 150,000 unique parent–child pairs, generating about 600,000 pair-wave observations across all survey years. I restrict the analysis to 1998–2018 because child income is categorized using different bins prior to 1998, making earlier waves incompatible with later ones, and because ending the sample in 2018 avoids distortions associated with the COVID-19 pandemic.

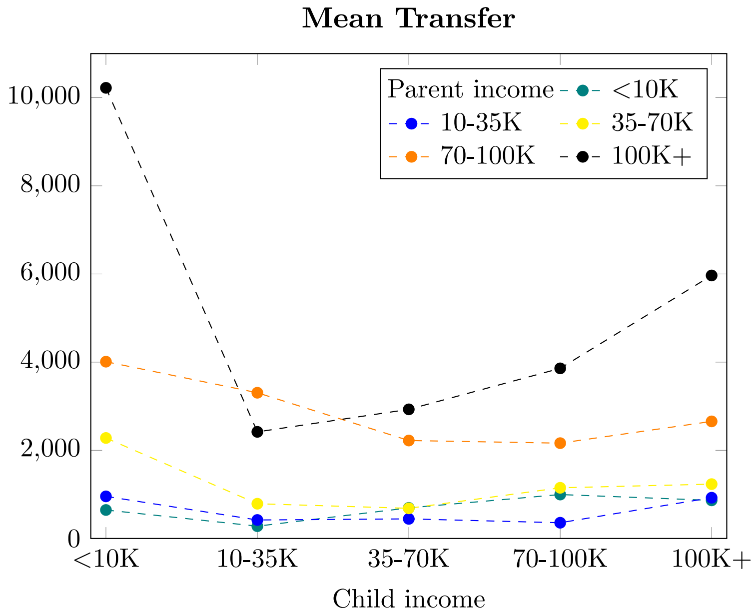
3.1 Transfers by Parent and Child Incomes

To evaluate the theory of altruism, I examine transfers in relation to parental and child resources, focusing on three objects: mean unconditional transfers, the extensive margin, and the intensive margin[1]. Figure 2 presents average transfers across parent and child income groups. Some parent income profiles display the expected pattern of declining transfers as child income rises, but others

[1]The extensive margin refers to the probability of a parent giving a transfer to a particular child. The intensive margin is the mean transfer given to children conditional on a parent giving a positive transfer.

do not. For several parent income groups the relationship is flat, and for the highest income parents it is increasing. These unconditional patterns suggest that a basic model of homogeneous altruism does not capture the full range of transfer behaviour observed in the data.

Figure 2: Unconditional Average Transfers by Parent and Child Incomes

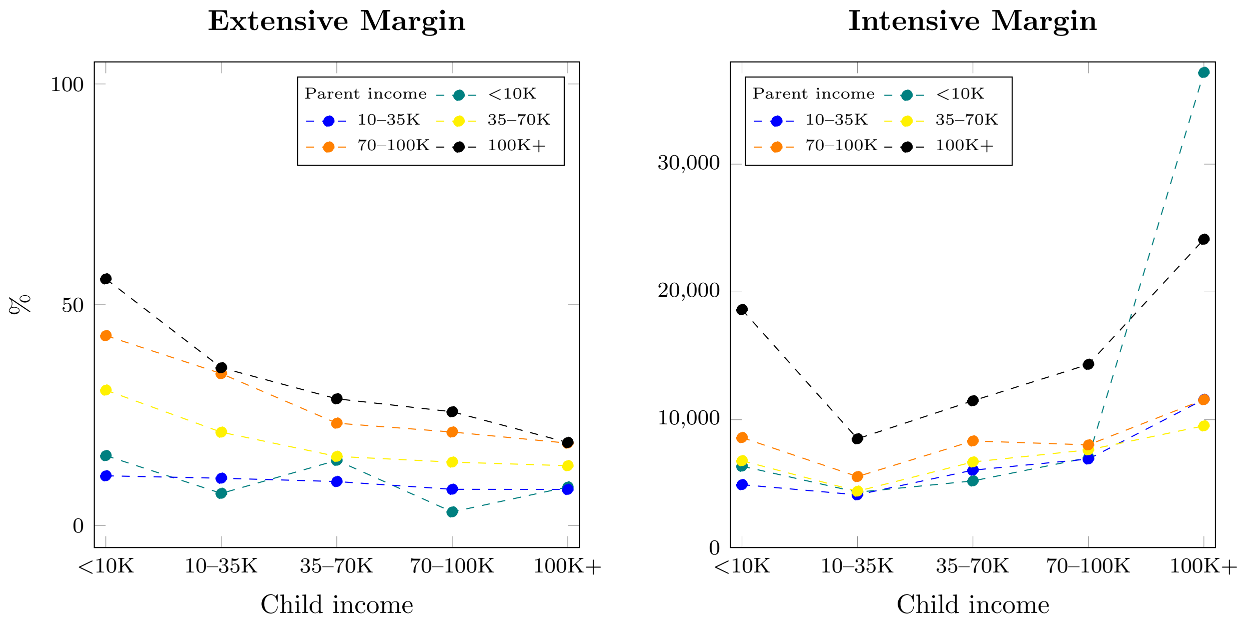


Note: Mean transfers from parents to children by common income categories. The horizontal axis represents child income, while parent incomes are differentiated by the five plotted curves. Transfers are given between 1998–2018, measured in two-year survey intervals, and dollar figures are expressed in 2018 terms.

Figure 3 shows that the extensive margin largely conforms with the standard theory. Higher income parents are more likely to give transfers, and the likelihood of giving declines as child income rises. This pattern is consistent with motives of intergenerational consumption smoothing, where parents respond to differences in need between themselves and their children.

The intensive margin is relatively stable across most parent and child income groups. This trend stands in sharp contrast to the predictions of standard theory: under a motive of homogeneous altruism, the intensive margin should be largest for low-income children and should decline steadily as child incomes rise. Instead, for most parent income groups, transfer amounts are relatively flat across child income bins until the very top of the distribution, where transfers rise again. This pattern is inconsistent with a needs-based view of inter vivos transfers and suggests that parents' reasons for giving are more complex than the standard assumption of shared utility and consumption smoothing.

Figure 3: Extensive and Intensive Margins



Note: The extensive margin is increasing in parent income and declining in child income, suggesting an altruism motive among parents that necessitates consumption smoothing with their child. In contrast, the intensive margin is primarily flat for most child incomes, with some variation among high income parents and notable increases in the intensive margin to the highest-income children.

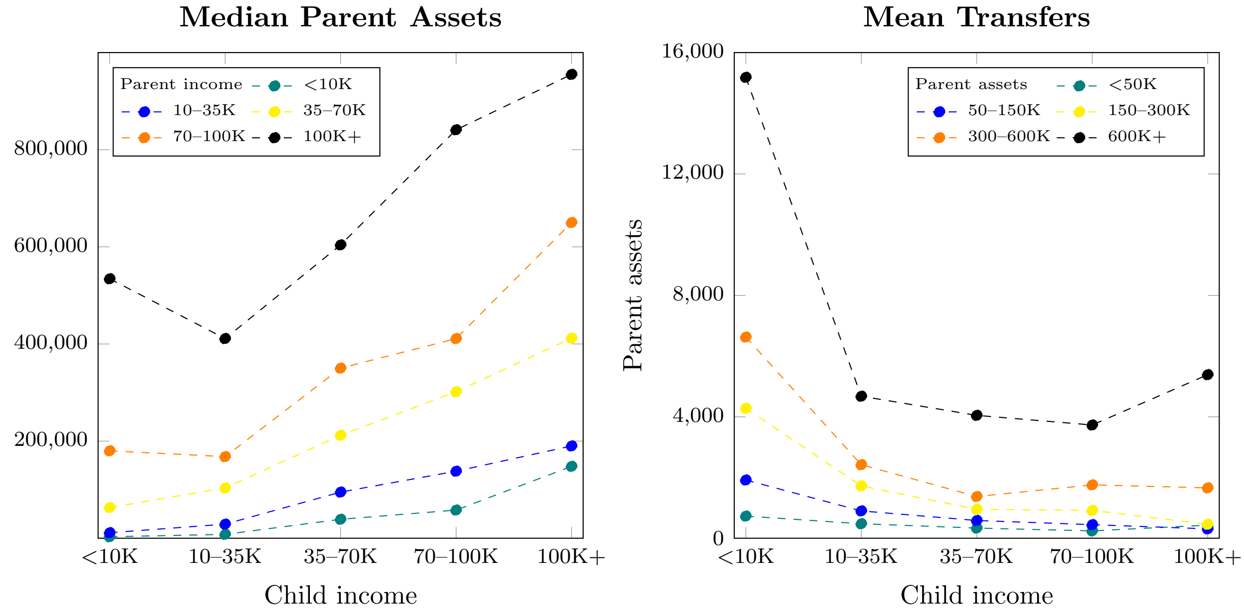
Three cases in the intensive margin plot of Figure 3 are particularly salient when considered in the context of altruism theory. First, a large intensive margin from the highest-income parents to the lowest-income children is consistent with consumption smoothing and joint utility maximization. Second, the highest-income parents also give large transfers (when they give) to high-income children. Although this behaviour is not well explained by homogeneous altruism, it may be indicative of a dynastic savings motive or even the timing of strategic bequests to children who are financially stable. Third, the lowest-income parents surprisingly also give large transfers to the highest-income children. To rationalize this behaviour, consider that transfers from parents may be paid out from income or assets, and as parents age they increasingly fund expenditures via dis-saving. The intensive margin is reflective of a mix of insurance, dynastic, and asset-driven motives that standard theory cannot fully account for.

A key fact in understanding increasing average transfers and the rising intensive margin for some high-income children is the relationship between parent assets and child income. Since parent and child incomes are positively correlated across generations, and parent incomes are themselves strongly correlated with accumulated assets, children with higher incomes tend to have parents with greater financial resources. The left panel of Figure 4 illustrates this intergenerational linkage: except for the lowest child-income bin, median parent assets rise monotonically with child income within every parent-income group.

This trend is consistent except among children with incomes below \$10,000, where the relationship between child income and parent assets is mixed: when children move to a slightly higher income bin (10–35K), assets increase slightly for low-income parents, remain flat for upper-middle-income parents, and decrease somewhat among the highest income parents. Beyond this lowest child income category, however, median parental assets are monotonically increasing in child incomes for all parents regardless of income.

The right panel of Figure 4 shows that mean transfers by parent assets follow a similar pattern as mean transfers by parent income in Figure 2. For each parent-asset group, transfers are relatively flat across most child-income categories, with pronounced increases only for the lowest-income and highest-income children. This similarity to Figure 2 is informative: when conditioning on parent wealth rather than parent income, the intensive margin still does not exhibit the declining pattern predicted by basic altruism theory. Parent assets, like parent incomes, are increasing with child income, and mean transfers do not fall either. Accounting for wealth therefore does not rectify the inconsistency between the data and the standard model. Instead, the joint distribution of parent

Figure 4: Parent Assets and Mean Transfers by Child Income



Note: The left panel shows median parent assets for each combination of parent and child incomes. Parent assets are increasing in child income except among the lowest-income children. The right panel details mean transfers by parent assets instead of income. Similar to Figure 2, transfers are relatively flat in child incomes with the exception of the lowest-income and highest-income children.

assets and child income reinforces the conclusion that richer motives are necessary to explain observed transfer patterns.

3.2 Regression Analysis: Transfer Amounts

Table 1 reports ordinary least squares coefficients for a variety of parent and child characteristics on transfer quantities from parents to adult children. This section will focus on the effects of child and parent incomes as well as parent assets on transfer amounts. For the full regression estimation, see appendix A.6.

Coefficients on child incomes show a clear and monotonic trend: relative to children with incomes below $10,000, parents give substantially smaller transfers to children with higher incomes. This decline is roughly equivalent in financial terms to a $2,500 reduction among children earning

\$10–35,000, up to a \$5,450 reduction in transfers for children earning above \$100,000. Consistent with motives of intergenerational consumption smoothing, parents give their largest transfers to children who earn the least. Further, this consumption smoothing mechanism is also reflected in the coefficient on children's full time work. Relative to those who are not working, adult children working full time receive approximately \$300 less from parents even after accounting for differences in child incomes.

Parent income coefficients are positive, statistically significant, and quantitatively relevant. As parent incomes rise, they increase transfers to their children by approximately 17 cents for every dollar of income. This aligns with the broader empirical pattern that financially stronger parents are more likely to give and give more. Additionally, parents who own their own homes provide slightly larger transfers, suggesting that financial and housing security is a key ingredient in parental support to children.

The effects of parent assets and primary residence value in the regression are statistically significant, even if numerically small. Comparing the magnitude of these coefficients relative to parent income suggests that while measures of wealth help explain the variation in transfers, they are not a major determinant of transfer amounts. For each additional \$100,000 in assets, parents increase their transfers by slightly more than three cents. At the same time, assuming income and wealth are substitutable for parents implies that the same increase of \$100,000 applied to parent income would increase transfers by approximately \$1,600.

A final implication of the regression analysis concerns the coefficient of determination, or the explainable share of variation in transfers. Controlling for a range of demographic characteristics, the leftmost specification in Table 1 accounts for only about 16 percent of the variation in transfer amounts. Further, controlling only for incomes and assets of parents and children explains just under 6 percent of the variation in transfer. In the context of existing theories of altruism, this R^2 is informative: the regression implicitly assumes that all parents share the same underlying degree of altruism, because altruism is not directly observable in the HRS and therefore cannot be included

Table 1: OLS Estimates for Transfer Amount

	All Covariates		**Income and Assets Only**	
Independent Variable	**Coefficient**	**p-value**	**Coefficient**	**p-value**
Child Income				
<10K	*Base*		*Base*	
10–35K	−0.0589	0.000	−0.0771	0.000
35–70K	−0.0829	0.000	−0.0968	0.000
70–100K	−0.0810	0.000	−0.0988	0.000
100K+	−0.1281	0.000	−0.1013	0.000
Parent Income and Assets				
Parent Income	0.0165	0.000	0.0230	0.000
Parent Assets	3.53×10^{-8}	0.000	4.07×10^{-8}	0.000
Parent Owns Home	0.0052	0.026		
Primary Residence Value	4.66×10^{-8}	0.000		
Child Employment				
Not Working	*Base*			
Part-time Work	0.0000	0.992		
Full-time Work	−0.0070	0.008		
Child Enrolled in School	0.0282	0.000		
Child Lives with Parents	0.0327	0.000		
Coefficient of Determination	0.1633		0.0587	

Note: Weighted ordinary least squares estimation using HRS data. The first specification includes a broad set of observables detailed in Appendix A.6. The second specification is limited to controlling for only child income, parent income, and parent assets. Coefficient values are scaled by median parent income ($42,600). Transfer values are reported over a two year interval.

as a covariate. The relatively weak model fit, even with many demographic characteristics, indicates that variation in transfers is not well accounted for by common observables. In the following section, I will propose a new mechanism, heterogeneity in parental altruism, to better account for transfer behaviour from parents to adult children.

4 Model

The model is static and endowment-based where parents are the sole decision-makers. They choose how much of their endowment (e_p) and current assets (a) to dedicate to consumption (c_p), savings (a'), and transfers to a child (t). The child's consumption (c_k) will be determined completely by their endowment (e_k) and the transfer they receive from their parent. A parent in this environment will equate the marginal utility of allocating an additional unit of their endowment between their consumption, their child's consumption (via transfer), and saving.

$$\max_{t,a'} \frac{c_p^{1-\sigma}}{1-\sigma} + \nu \frac{c_k^{1-\sigma}}{1-\sigma} + \frac{\psi_1(\psi_2 + a')^{1-\sigma}}{1-\sigma} \tag{2}$$

subject to

$$\begin{aligned} c_p &= e_p + a - a' - t \\ c_k &= e_k + t \\ t &\geq 0 \\ a' &\geq 0 \end{aligned}$$

The first two terms in equation (2) refer to utility from both their own and their child's consumption. The third term is a warm-glow function adapted from Jones and Li (2023) which represents utility from bequests. Since this environment is static, savings do not serve as precautionary self-insurance like they would in a life cycle setting. Instead, I assume parents allocate resources to savings as a bequest that will transfer to their children at some point in the future.

Heterogeneity in parent and child endowments allows the model to account for basic altruism theory at the bottom end of child incomes. As parent incomes rise, transfers are also increasing; in contrast, child incomes are negatively related to the quantity and likelihood of transfers. Introducing variation in parental altruism improves the model's ability to match these patterns by producing positive transfers to children throughout the income distribution. My preferred calibration assumes that altruism follows a lognormal distribution, though alternative specifications are also explored.

$$\ln \nu \sim N(\mu_\nu, \sigma_\nu) \tag{3}$$

5 Calibration

Parameters external to the model include risk aversion (σ), a transfer cutoff (t_ℓ) to mimic the survey design of the HRS, and primitives for the warm-glow functional form (ψ_1, ψ_2). The transfer cutoff approximates restrictions on the HRS transfer questions which report only amounts above $500 over two years. Warm-glow parameters approximate the overall preference for bequests among donors (ψ_1) as well as non-linearity or the degree to which bequests are seen as a luxury good (ψ_2).

Table 2: External Parameters

Parameter	**Description**	**Value**	**Source**
σ	Risk aversion	2	Literature
t_ℓ	Transfer threshold	0.008195	Data
ψ_1	Saving preference	2.726	Jones and Li (2023)
ψ_2	Saving non-linearity	13.4	Jones and Li (2023)

The altruism distribution for parents is defined by two internally calibrated parameters. Altruism is distributed according to a lognormal distribution, where its mean (μ_ν) is calibrated to match mean annualized transfers from the HRS, scaled to median parent income of $42,600. Second, the variance parameter (σ_ν) is targeted to match the variation in transfers (R^2) explained by standard demographic observables. The approach is to calibrate altruism variance to capture all explained variation in the data rather than have an expansive model with many observables.

Table 3: Internal Parameters

Parameter	**Description**	**Value**	**Target**	**Model**	**Data**
μ_ν	Mean of altruism distribution	-8.10778	Average Transfer	0.0375	0.0375
σ_ν	Variance of altruism distribution	0.25919	Explained Variation	0.1933	0.1933

In addition to the explicitly calibrated parameters, the model directly incorporates the empirical joint distribution of child income, parent income, and parent assets from the HRS. These distributions enter as endowments corresponding to each parent–child pair, ensuring that the model is evaluated over the same heterogeneity in resources observed in the data. Importing the full joint distribution allows the model to preserve the empirical correlations between income and wealth across generations and provides a foundation for assessing how variation in altruism interacts with transfer flows from parents to children.

6 Results

This section presents the main quantitative results of the model and evaluates its ability to replicate the empirical relationship between child incomes, parent incomes, and transfers. I show how incorporating heterogeneity in parental altruism as well as the joint distribution of parent and child endowments generates positive transfers across the income distribution, including to some higher-income children for whom homogeneous altruism does not generate transfers. I also test alternative specifications for parental altruism beyond the model assumption of a lognormal distribution.

6.1 Model Validation

The model is evaluated using an ordinary least squares regression on transfer amounts from parents to adult children. By comparing regression estimations across the HRS as well as the calibrated model, I can account for variation explained by transfers as well as unexplained variation beyond the model states of parent and child endowments. The first panel of Table 4 show the coefficient of variation on transfers (CV_{Tr}) among the HRS data, a standard model with homogeneous altruism targeted to mean transfers, and the calibrated model with lognormally distributed parental altruism.

Table 4: OLS Comparison of Homogeneous and Heterogeneous Altruism Models

	Data	**Homogeneous**	**Heterogeneous**
Panel A: Measure of Variation			
CV_{Tr}	0.1603	0.3012	0.1593
Panel B: Regression Coefficients			
10–35K	−0.0589	−0.4324	−0.3223
	(0.0027)	(0.0058)	(0.0505)
35–70K	−0.0829	−0.4218	−0.3287
	(0.0033)	(0.0067)	(0.0587)
70–100K	−0.0810	−0.4082	−0.3155
	(0.0044)	(0.0093)	(0.0817)
100K+	−0.1281	−0.3751	−0.2790
	(0.0057)	(0.0156)	(0.1368)

Note: Panel A reports measures of variation in transfers across the data and model specifications. Panel B reports regression coefficients for child income bins relative to children with income below \$10,000. Transfer amounts are scaled by median parent income (\$42,600).

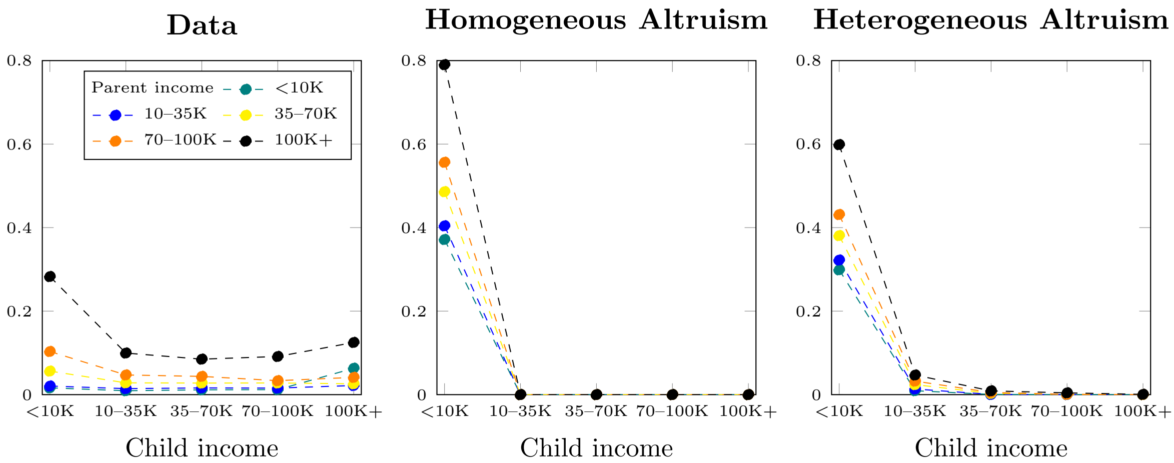
Aside from the aforementioned measures of variation, Table 4 also summarizes the relationship between child incomes and transfers in the data as well as both model specifications. Following the structure of the HRS, child incomes are sorted into bins and regression coefficients represent the average absolute change in mean transfer size relative to an otherwise identical child with income below \$10,000. Moving from a model of homogeneous altruism to one with heterogeneous altruism reduces the relative effect of child incomes on transfers; this is consistent with adding an additional source of variation via the altruism distribution.

Although the heterogeneous altruism model improves the fit of average transfers relative to the homogeneous benchmark, the coefficients in Panel B remain larger in magnitude than those observed in the HRS.

6.2 Model Transfers from Parents to Children

Average financial transfers from parents to adult children in the data are characterized by increasing quantities in parent incomes and decreasing quantities in child incomes. Further, positive transfers occur across the joint distribution of parent and child incomes. A theory of homogeneous altruism can account for the first feature, but not the second. As shown in Figure 5, a benchmark homogeneous model calibrated to mean transfers results in all transfers going towards children with the lowest possible income (<10K). Mechanically, this arises because for parents to give transfers to higher-income children, they require a level of altruism that would result in an over-estimation of transfers in the aggregate. When targeting mean transfers from parents who are identically altruistic, the observed flows from the model will also be targeted to the poorest children with the most severe need.

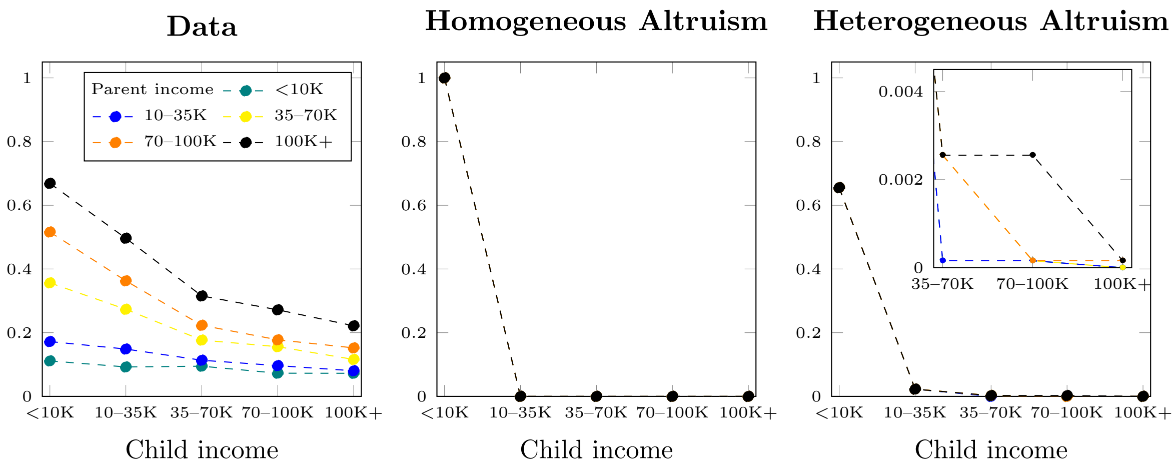
Figure 5: Average Transfers



Note: Unconditional mean transfers by parent and child incomes. Amounts are scaled by median parent income. Allowing for heterogeneous altruism generates positive transfers over a longer range of child incomes to more closely approximate the data.

Accounting for heterogeneity in parental altruism allows the model to generate positive transfers to some higher-income children, not only those at the bottom of the income distribution. In the calibrated model, transfers flow to low-income children and also to children at higher income levels, particularly when parents draw sufficiently high altruism. These transfers remain quantitatively small on average because they are made by a relatively narrow subset of highly altruistic parents. This upward extension of transfers necessarily reduces transfers to the lowest-income children, since the model targets average transfers; any resources directed toward higher-income children must come from the transfers that, under homogeneous altruism, would have gone exclusively to the poorest children.

Figure 6: Extensive Margin

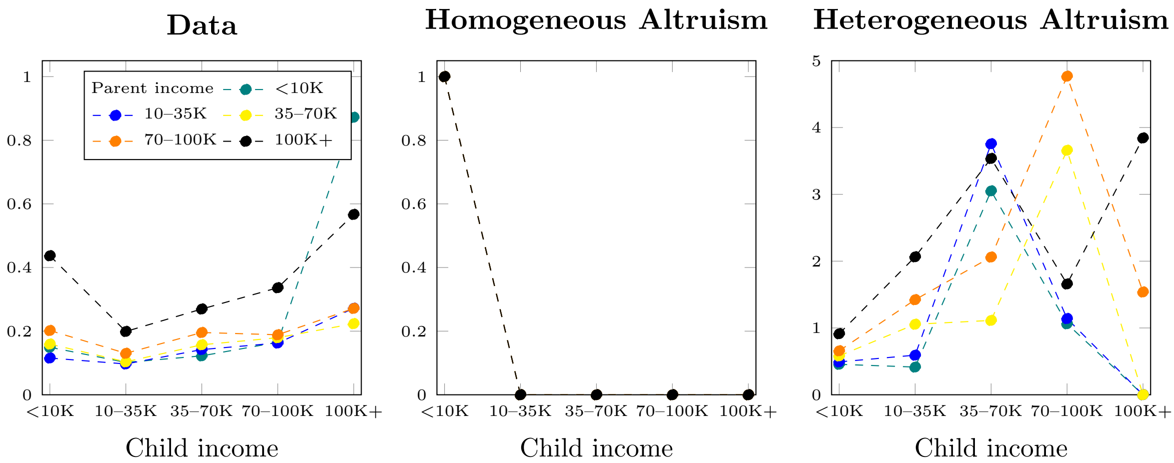


Note: Probability of transfer for each parent–child pair, grouped by common income categories. A model with heterogeneous altruism generates positive transfers among higher-income children. In contrast, in a homogeneous altruism framework all transfers cluster towards the poorest children only.

The extensive margin follows a similar trend relative to mean transfers. In the data, transfers are at least somewhat common across all parent and child incomes, and are very common among low-income children. In both model specifications, the extensive margin is a non-targeted moment and will reflect the changes in mean transfers discussed previously. Specifically, moving to a model of heterogeneous altruism results in a positive extensive margin at all child income levels, and a reduction in estimates for the extensive margin among lowest-income children relative to a model of homogeneous altruism.

In the data, for parents at any income level and for children above the lowest income bin, the intensive margin of transfers increases with child incomes, and the rise is especially pronounced among the highest-income children (100K+). This runs counter to motives of intergenerational consumption smoothing and represents a significant departure from the existing understanding of parental altruism. Under a theory of homogeneous altruism where heterogeneity is only incorporated via parent and child resources, each pair of parents and children will give (or not give) a transfer with certainty. The mechanical result of this is that the intensive margin under homogeneous altruism is identical to unconditional mean transfers.

Figure 7: Intensive Margin



Note: Transfer amounts among those who give or receive a transfer. A model of heterogeneous altruism can generate an increasing intensive margin among some parent and child incomes, however in levels it is an overestimate and the relationship between the intensive margin and income is extremely noisy.

However, imposing heterogeneous altruism allows for variation in transfers within pairs of parent and child incomes. This change has two effects: first, quantitative estimates for the intensive margin are very large relative to the data. While model transfers themselves are targeted, the intensive margin is weighted by the inverse of the mass of parents who give transfers. Since the mass of extremely high altruism parents is low, estimates for the intensive margin become very large. Second, there is some positivity in the intensive margin with respect to child incomes. However, as a result of the large variation in and over-estimation of the intensive margin, heterogeneous altruism does not provide a clear explanation for this empirical pattern.

Introducing heterogeneous altruism improves the model's ability to generate transfers to children across the income distribution. This is most visible in the extensive margin, where a small share of highly altruistic parents make transfers to almost all children regardless of income. This expansion in the extensive margin also has implications for average transfers, reducing the relative density among the lowest-income children to more closely match transfers in the data. However, this small share of highly-altruistic parents also inflate the intensive margin far beyond values observed in the data, although there is some positivity in child incomes that is absent from the homogeneous baseline. The main contribution of heterogeneous altruism in this context is therefore its ability to broaden the set of parent–child pairs who give transfers, rather than to refine the magnitude of transfers among those who do.

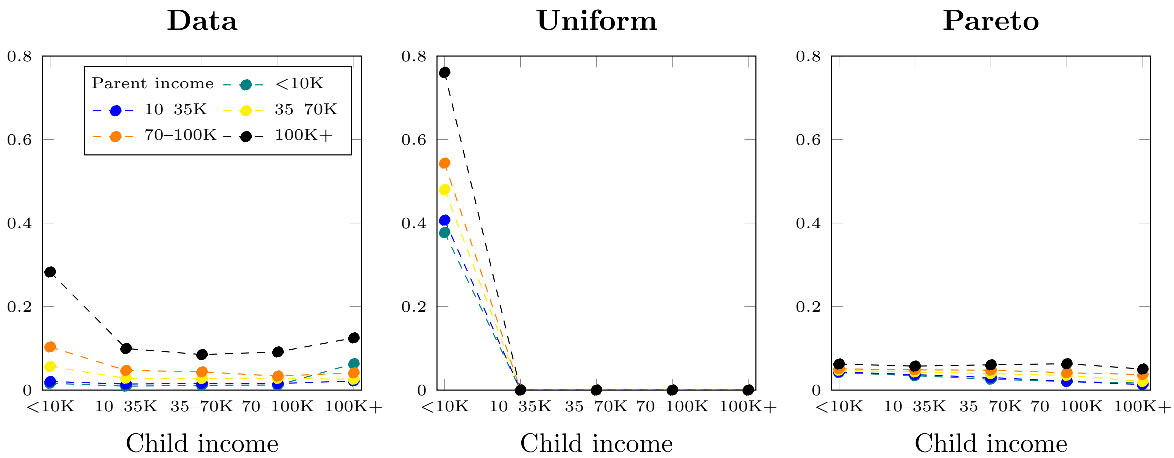
Table 5: Parameters – Alternative Altruism Specifications

Parameter		Value	Target	Model	Data
Panel A: Uniform Distribution					
$\max_\nu$	Maximum Altruism	0.0021	Average Transfer	0.0375	0.0375
Panel B: Pareto Distribution					
α	Shape Parameter	0.8885	Average Transfer	0.0375	0.0375
x_m	Scale Parameter	0.0001			
$\max_\nu$	Maximum Altruism	1			

6.3 Alternative Specifications for Parental Altruism

The previous results rely on the use of a lognormal distribution for parental altruism. To test this, I present model moments under heterogeneous altruism with uniform and Pareto distributions in lieu of the lognormal specification discussed previously. Mean transfers under a uniform distribution as shown in Figure 8 resemble the previous homogeneous case: this is because the maximum level of altruism is calibrated to match average transfers, and its minimum is bounded at zero. Since there are an equal mass of parents across the altruism distribution, the distribution's upper bound must by necessity be lower than under a lognormal distribution where the masses of parents are skewed towards the left tail of the distribution. The result is that no parents are as altruistic as they would be under a non-uniform or right-skewed distribution, and the model cannot generate transfers to higher-income children.

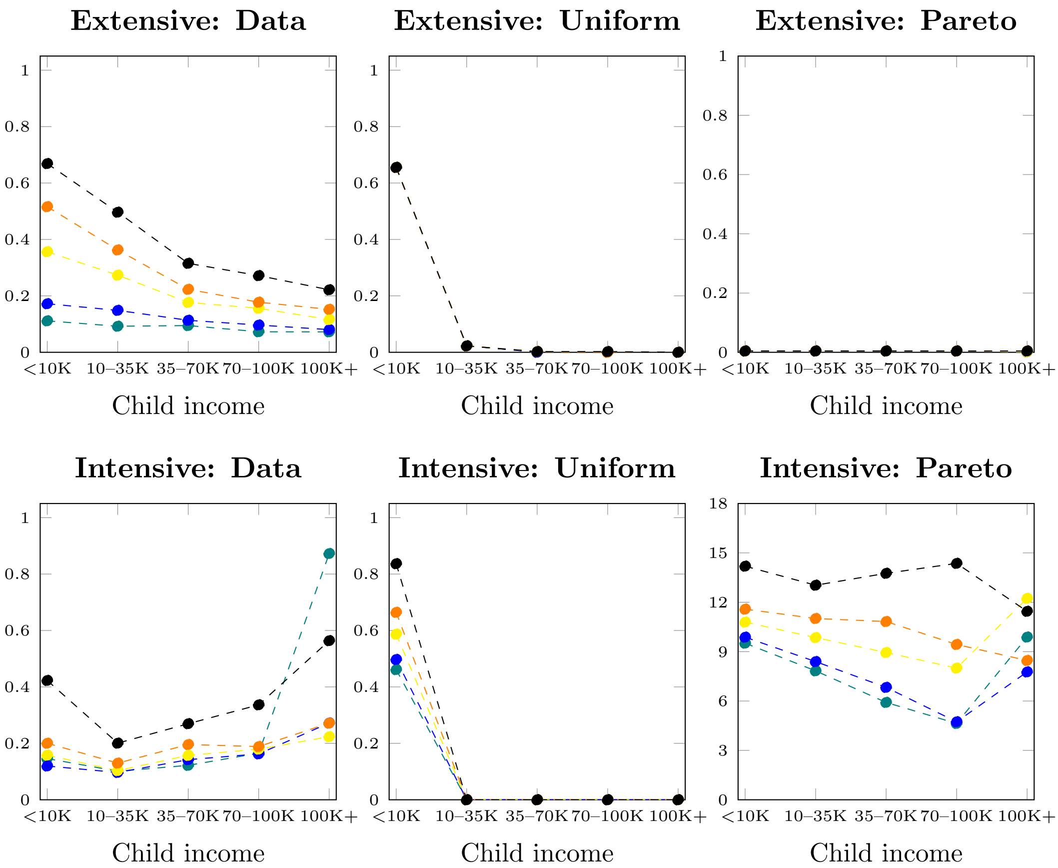
Figure 8: Average Transfers Under Uniform and Pareto-Distributed Altruism



Note: Average transfers by parent and child incomes under alternative altruism distributions. Under a uniform distribution, the upper bound on altruism is calibrated to match mean transfers, resulting in limited variation and no transfers to higher-income children. A Pareto distribution captures the left-skew and heavy right tail of altruism, generating positive transfers across all child incomes but with less variation than the lognormal benchmark.

Employing a Pareto distribution accounts for the extreme right tail behaviour present in a lognormal distribution. On parametrization, the scale parameter (x_m) is set by assumption as a small positive number and the shape parameter (α) is targeted to average mean unconditional transfers. Further, I set an upper bound on the Pareto distribution at $\nu = 1$, under the assumption that no parent cares more for their child's utility than their own. This model specification generates positive transfers across all child incomes, but with much lower variation than in the lognormal case. The source of this lower variation can be seen in the extensive margin panel of Figure 9: under a Pareto distribution, very high altruism parents give transfers regardless of their or their child's income. This results in an equilibrium where a small number of parents always give transfers but the vast majority always give nothing.

Figure 9: Extensive and Intensive Margins Under Uniform and Pareto-Distributed Altruism



Note: Extensive and intensive margins of transfers under uniform and Pareto-distributed altruism. A uniform distribution produces transfer patterns similar to the homogeneous altruism model, with transfers concentrated among the lowest-income children. A Pareto distribution generates positive transfers across all child incomes, but most parents give nothing while a small number of highly altruistic parents give regardless of income. This produces limited variation in the extensive margin and large, noisy estimates of the intensive margin.

7 Conclusion

This paper shows that a standard theory of homogeneous altruism cannot account for observed inter vivos transfers, even after conditioning on the empirical distributions of parent income, child income, and parental wealth. Using linked longitudinal data from the Health and Retirement Study, I document the empirical patterns of transfer behavior between parents and adult children. The evidence is consistent with a basic motive of consumption smoothing: parents give larger transfers to children with lower incomes and to those out of work, even after conditioning on a rich set of covariates. At the same time, substantial transfers are made to higher-income children, indicating that observable economic characteristics alone cannot fully explain the distribution of transfers.

To evaluate the fit of standard altruism theory to observed transfers, I estimate a regression of transfer amounts on parent income, child income, parental assets, and a broad set of demographic controls. This estimation can explain around 16 percent of the variation in transfers. I interpret the unexplained variation as reflecting unobserved heterogeneity in parental altruism and use this insight to develop the structure of the model.

I construct and estimate a static model of heterogeneous altruism in which parents differ in the strength of their altruistic preferences. Allowing altruism to vary across households enables the model to generate transfer patterns that more closely resemble those in the data, including positive transfers to some higher-income children that homogeneous altruism models cannot reproduce.

These findings carry important implications for the design and evaluation of public policy. Standard models of homogeneous parental altruism imply that private transfers flow primarily to financially constrained children. The evidence presented here shows that this theory is incomplete, since substantial transfers also flow to higher-income children who have relatively little need for parental support. Applying the canonical model in policy contexts will lead to systematic underestimation of private transfers to high-income individuals and overestimation of family support to low-income individuals. Incorporating heterogeneity in altruism would improve the targeting of benefits, refine assessments of crowd-out, and yield more accurate predictions of the distributional impacts of government policy.

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A Appendix

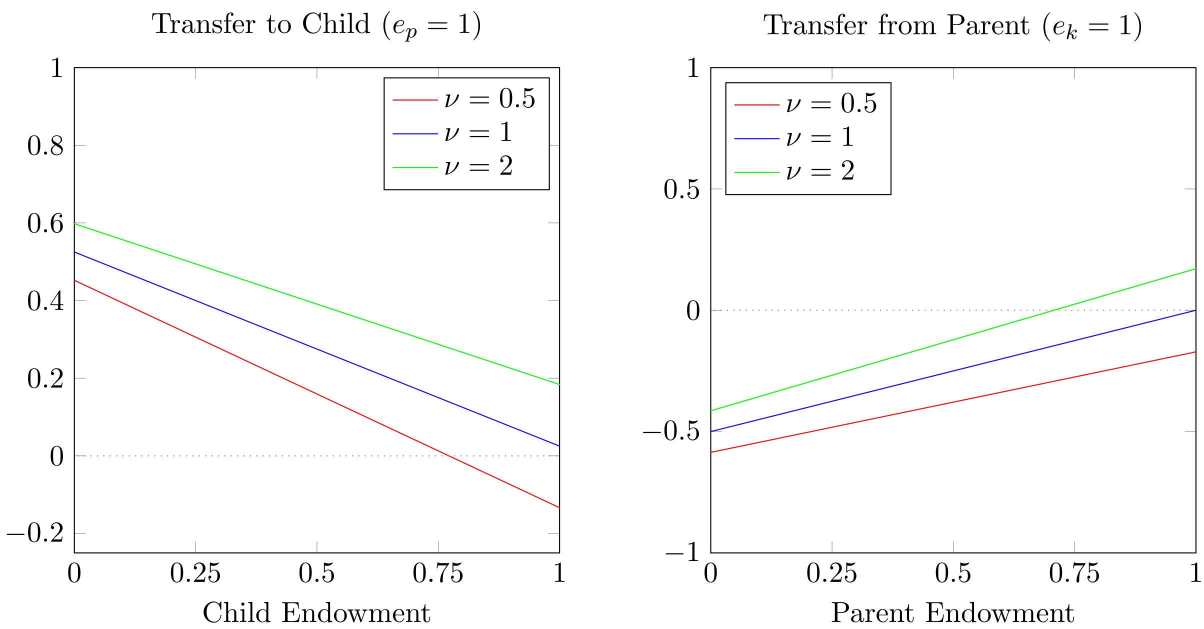
A.1 Heterogeneity in Subsistence Consumption

The baseline model of altruistic parents assumes that all consumption is valued symmetrically, but in many real-world settings parents and children face different subsistence needs. A retired parent living in a low-cost area may require relatively little to meet basic needs, while a young adult child renting in a high-cost city may face a much higher subsistence threshold. This can be captured by introducing $\bar{c_p}$ and $\bar{c_k}$, the minimum consumption levels required by parents and children. Incorporating these thresholds modifies the transfer function to:

$$t^* = \frac{\nu^{1/\sigma}(e_p - \bar{c}_p) - (e_k - \bar{c}_k)}{1 + \nu^{1/\sigma}}$$

Figure 2 shows how these subsistence levels shift equilibrium transfers. When children face higher subsistence needs than parents, such as when $\bar{c_k} > \bar{c_p}$, transfers shift upward across all

Figure 10: Transfer Functions with Heterogeneous Subsistence Consumption



Note: Imposing heterogeneity in subsistence consumption induces a small positive shift in transfers across the range of parent and child endowments. This occurs because parents face a lower subsistence requirement relative to their children.

levels of altruism. Intuitively, children require more support to reach the same effective level of discretionary consumption. The opposite case is equally informative; if parents face higher subsistence needs, perhaps as a function of health shocks, then $\bar{c_k} < \bar{c_p}$ and transfers shift downward. Parents have less post-subsistence income available to redistribute, even if their degree of altruism remains unchanged.

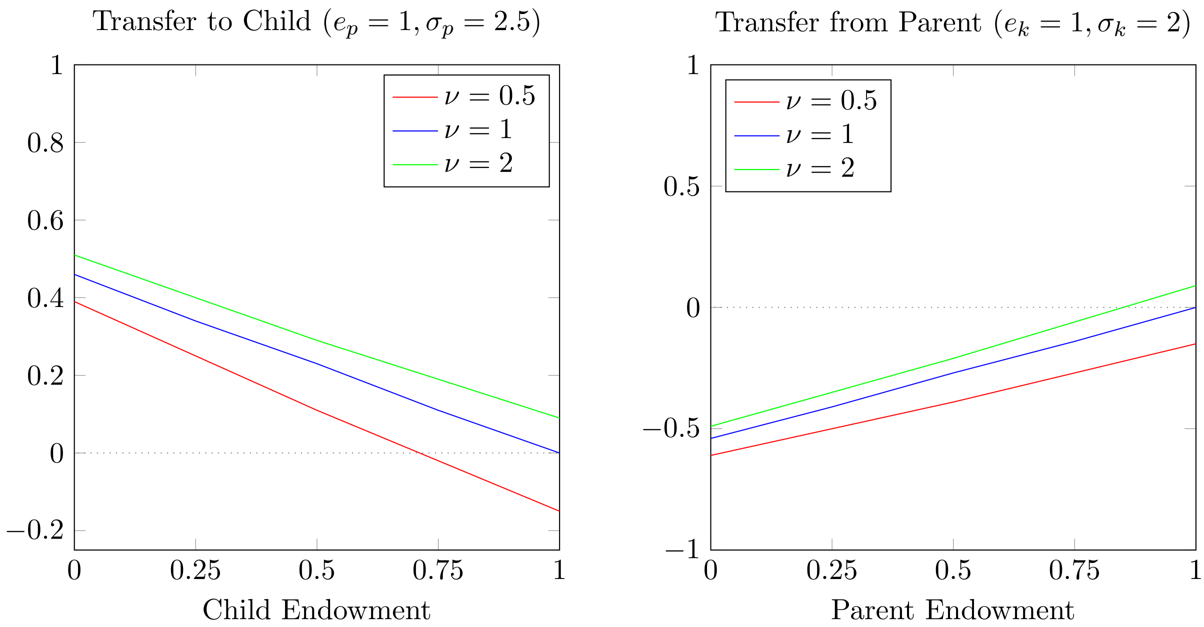
A.2 Heterogeneity in Risk Aversion

An additional extension to the simple model of parental altruism allows parents and children to differ in their degree of risk aversion. This is particularly relevant in settings where parents are older and more financially conservative, while children are younger and relatively less risk-averse. Suppose, for example, that parents have $\sigma_p = 2.5$ and children have $\sigma_k = 2$. The equilibrium condition becomes:

$$\nu = \frac{(e_k + t)^{\sigma_k}}{(e_p - t)^{\sigma_p}}$$

which must be solved numerically.

Figure 11: Transfer Functions with Heterogeneous Risk Aversion



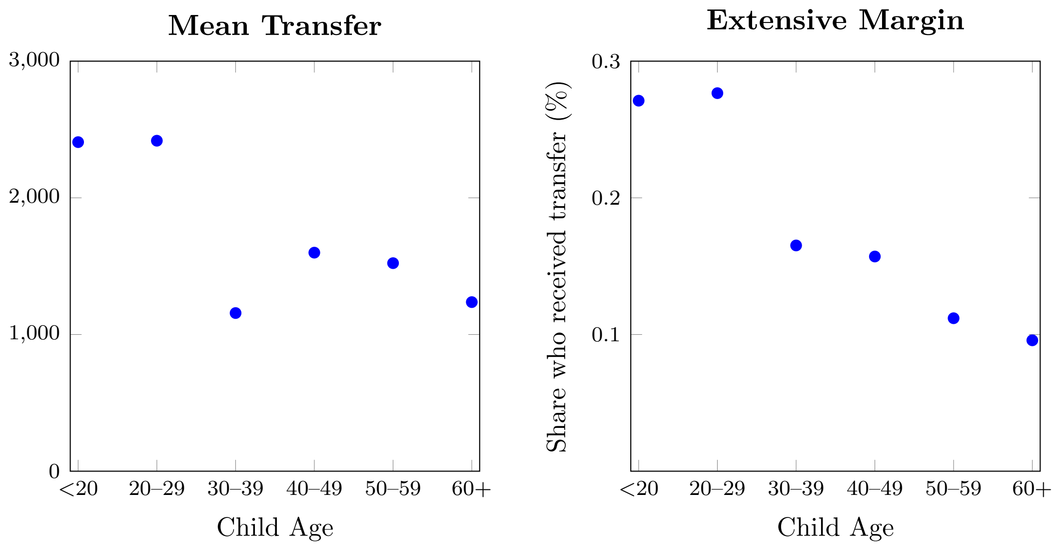
Note: Imposing heterogeneity in risk aversion generates a reduction in the slope of the transfer functions when $\nu = 2$ and $\nu = 0.5$. This results in a tighter dispersion of transfers relative to parental altruism, reflecting the fact that more risk-averse parents give larger transfers.

Figure 3 details equilibrium transfers under this new form of heterogeneity. The main effect is a flattening of the transfer curves for both low and high altruism ($\nu = .5$ and $\nu = 2$). When parents are more risk averse, they place greater value on smoothing consumption across states, and this leads them to give larger transfers regardless of their altruism weight. The dispersion of transfers across different values of ν narrows, reflecting the dominant role of risk preferences in shaping the marginal utility tradeoff.

A.3 Transfers Across the Life Cycle

Standard theory and existing empirical work emphasize that parental transfers are most important in early adulthood. Many of the largest lifetime expenditures such as post-secondary education, first-time home purchases, and other major investments occur when incomes and asset accumulation are lowest. Intuitively, these expenses are often at least partially financed by parents, and indeed average transfers are highest among young adults.

Figure 12: Transfer Flows Across the Life Cycle



Note: Unconditional average transfers and the probability of receiving a transfer by child age group. Transfers are highest among young adults, reflecting support for major early-life expenses, but remain significant throughout the life cycle. Even among children aged 60 and over with at least one living parent, approximately 10% receive a transfer from their parent. Transfers are in 2018 dollars and are measured cumulatively over the two-year interval between survey waves.

Figure 7 summarizes how transfers evolve across the child's life cycle, showing both unconditional average transfers and the probability of receiving a transfer. While young adults receive the largest transfers on average, financial support remains meaningful well into later adulthood. Even among children aged 60 and over with at least one living parent, approximately 10 percent receive a transfer, and average transfer amounts remain non-trivial.

Although transfers to older children are common, the motivations of donor parents likely shift as both generations age. Instead of financing large early-life expenditures, late-life transfers may compensate adult children for eldercare or other informal labour provided to aging parents. These transfers are far from rare: approximately 79 percent of children who receive any transfer are over age 30, underscoring that intergenerational financial support persists long after the early-career period typically emphasized in the literature.

A.4 Transfers Across Time

Across the length of the HRS sample, transfer behaviour from parents to adult children is remarkably stable. The share of parent–child pairs who engage in transfers in a given survey wave averages roughly 20 percent across all waves, with little systematic variation over time.

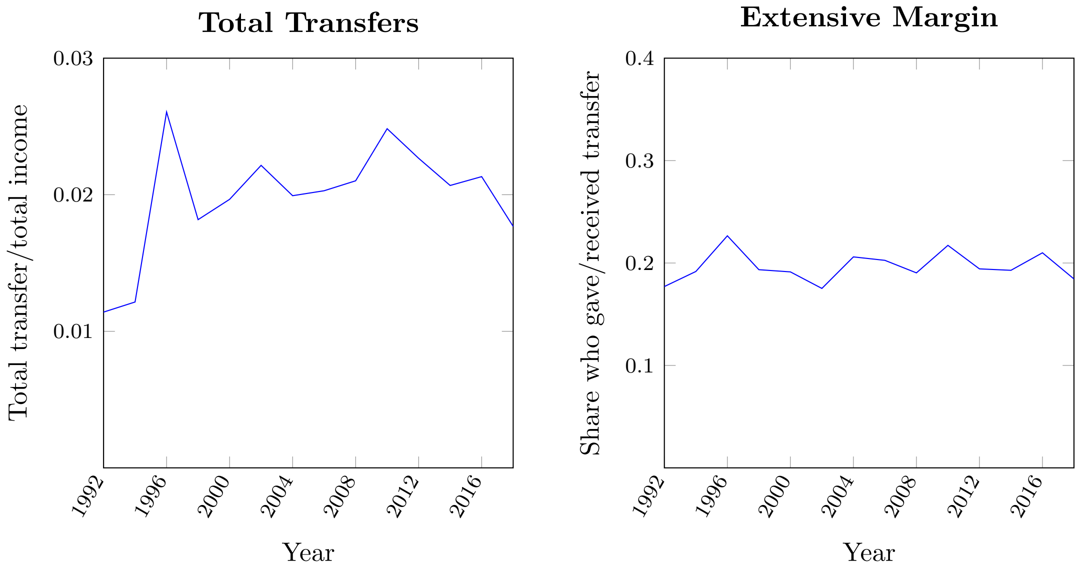
To quantify the real value of transfers, Figure 8 plots the ratio of total transfers to total parent income in each wave. In the early 1990s, this ratio exhibits substantial volatility, particularly around 1996. Part of this variation likely reflects changes in survey design during this period: between 1992 and 1996, the HRS sample was combined with the Asset and Health Dynamics Among the Oldest Old (AHEAD) survey, collected in 1993 and 1995. In the core sample from 1998 onward, the transfer-to-income ratio stabilizes at roughly 2 percent of parent income.

Relative to other household-level or government-level expenditures, this 2 percent figure is quantitatively meaningful. Over the same period (1998–2018), U.S. spending on unemployment insurance ranged from 0.2 to 0.5 percent of income, with a temporary increase to 1 percent following the 2008 financial crisis OECD, 2024. The stability and magnitude of parental transfers over time underscore their importance as a persistent component of intergenerational financial support.

A.5 Transfer Persistence

Transfers from parents to adult children are relatively persistent across survey waves. Roughly 40 percent of parents give two or more transfers to the same child, indicating that intergenerational

Figure 13: Transfer Flows Over Time



Note: The extensive margin of transfers remains stable at roughly 20 percent across the full HRS sample period. Total transfers as a share of total parent income exhibit volatility in the early 1990s but are relatively stable after 1998, averaging around 2 percent of parent incomes.

support is not a one-time event but an ongoing financial relationship for many families.

Parents who give repeated transfers tend to have higher incomes, greater asset holdings, and better health than parents who give few or no transfers. These patterns suggest that sustained giving is concentrated among households with both the resources and the stability to provide ongoing support. Transfer amounts also increase with the number of transfers given. Parents who give multiple transfers provide larger cumulative amounts: for instance, parents who give two transfers average $1,768, while those who give five transfers average $3,179. Persistence therefore reflects both the frequency and the magnitude of giving.

Table 1 highlights significant increases in both the frequency and size of transfers by parental economic status. Parents who give repeated transfers have substantially higher incomes, are much wealthier, and provide larger average transfers. For example, parents who give five transfers have average assets of roughly $567,000 and average transfers of $3,179, compared with $227,000 in

Table 6: Transfer Persistence and Parent Characteristics

# Transfers	Frequency	Mean Transfer	Parent Income	Parent Assets
0	11,797	0	43,629	227,109
1	5,079	755	63,254	315,845
2	3,590	1,768	77,498	380,087
3	2,336	2,069	89,079	549,182
4	1,734	3,380	93,431	489,430
5	1,219	3,179	93,392	567,385
6	975	4,383	101,961	721,359
7	686	3,883	101,250	702,017
8	552	4,777	122,460	849,300
9	393	5,909	129,122	877,015
10	293	6,135	125,419	936,660

Note: Each row corresponds to a parent–child pair grouped by the total number of transfers observed across survey waves. Parents who give repeated transfers tend to have higher incomes and larger asset holdings, and they provide larger average transfers. All monetary values are in 2018 dollars.

assets among parents who never give. Transfer persistence is thus concentrated among financially stronger parents, and children of these parents receive both more frequent and larger transfers across survey waves.

A.6 Regression: Transfer Amounts on Parent and Child Characteristics

Table 7 reports the full OLS specification first presented in Section 3.2. I estimate a regression equation of the form:

$$TransferAmount = \beta_0 + \boldsymbol{\beta_1}\ \boldsymbol{X^c} + \boldsymbol{\beta_2}\ \boldsymbol{X^p} + \boldsymbol{\beta_3}\ \boldsymbol{X^h} + \boldsymbol{\beta_4}\ \boldsymbol{X^e}$$

The vectors of characteristics ($\boldsymbol{X}$) are as follows:

- Child characteristics: $\boldsymbol{X^c}$ = Income, education, school enrollment, age, gender, employment status, marital status, parental status (number of children), co-Residence with parents
- Parent characteristics: $\boldsymbol{X^p}$ = Income, assets, primary residence value, age, education, em-

ployment status

- Parent health characteristics: $\boldsymbol{X^h}$ = High blood pressure, diabetes, lung disease, heart disease, ever had stroke, ever had cancer
- Eldercare characteristics: $\boldsymbol{X^e}$ = Child helps with ADLs, child helps with IADLs, child intends to help in future

Table 7: OLS Prediction for Transfer Amount

Independent Variable	**Coefficient**	**p-value**
Child Income		
<10K	*Base*	
10–35K	−0.0589	0.000
35–70K	−0.0829	0.000
70–100K	−0.0810	0.000
100K+	−0.1281	0.000
Parent Income & Assets		
Parent Income	0.0165	0.000
Parent Assets	3.53×10^{-8}	0.000
Parent Owns Home	0.0052	0.026
Primary Residence Value	4.66×10^{-8}	0.000
Child Age		
20–24	*Base*	
25–29	−0.0656	0.000
30–34	−0.0694	0.000
35–39	−0.0729	0.000
40–44	−0.0670	0.000
45–49	−0.0672	0.000
50–54	−0.0713	0.000
55–59	−0.0801	0.000
60–65	−0.0957	0.000
Child Demographics		
Child Gender (Female)	−0.0023	0.193
Child Marital Status	0.0027	0.068
Child Number of Own Children		

Independent Variable	**Coefficient**	**p-value**
0	*Base*	
1	0.0105	0.000
2	0.0119	0.000
3	0.0231	0.000
4	0.0284	0.000
5	0.0198	0.020
6	0.0348	0.010
7	0.0290	0.113
8	0.0282	0.395
9	0.0027	0.940
10	0.0192	0.782
11	0.0247	0.837
12	−0.0215	0.919
13	−0.0309	0.933
Child Years of Education		
0	*Base*	
1	0.0479	0.328
2	0.1288	0.017
3	0.0225	0.653
4	0.0671	0.164
5	0.0543	0.289
6	0.0894	0.008
7	0.0761	0.038
8	0.0606	0.035
9	0.0769	0.004
10	0.0740	0.005
11	0.0841	0.001
12	0.0698	0.006
13	0.0625	0.014
14	0.0854	0.001
15	0.1070	0.000
16	0.0890	0.000
17+	0.1026	0.000
Child Enrolled in School	0.0282	0.000
Child Employment		
Not Working	*Base*	
Part-time Work	0.0000	0.992
Full-time Work	−0.0070	0.008

Independent Variable	Coefficient	p-value
Child Lives with Parents	0.0327	0.000
Parent Age		
40–44	*Base*	
45–49	−0.0222	0.202
50–54	−0.0157	0.355
55–59	−0.0025	0.881
60–64	0.0139	0.413
65–69	0.0016	0.924
70–74	0.0201	0.248
75–79	0.0134	0.459
80–85	0.0529	0.006
Parent Marital Status	0.0001	0.977
Parent Number of Children		
1	*Base*	
2	−0.0585	0.000
3	−0.0850	0.000
4	−0.1111	0.000
5	−0.1248	0.000
6	−0.1129	0.000
7	−0.1074	0.000
8	−0.1119	0.000
9	−0.1269	0.000
10	−0.1073	0.000
11	−0.1035	0.000
Parent Education		
Less Than High School	*Base*	
GED	−0.0039	0.559
HS Graduate	−0.0005	0.909
Some College	−0.0114	0.011
College and Above	0.0310	0.000
Parent Employment		
Not working	*Base*	
Part-time Work	−0.0015	0.707
Full-time Work	−0.0022	0.416
Parent Health Limits Work	0.0032	0.100
Child Transfers to Parent	−0.0259	0.001

Independent Variable	Coefficient	p-value
Child ADL Help	−0.0284	0.048
Child IADL Help	−0.0124	0.336
Constant	0.0758	0.016

Note: Weighted ordinary least squares estimation using HRS data. Coefficient values are scaled by median parent income ($42,600). $R^2 = 0.1633, F = 209.58$